Technology Stack

Date	24 June 2026
Team ID	
Project Name	Human development index (HDI) Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML5, CSS3, Bootstrap, JavaScript	Used to create a responsive, user-friendly web interface for entering HDI indicators and displaying prediction results. Bootstrap ensures mobilefriendly and consistent UI design

2	**Backend / Server-Side**	Python, Flask	Python is the primary programming language for implementing the machine learning model, while Flask provides a lightweight web framework to connect the frontend with the prediction model efficiently.
3	**Database / Data Storage**	CSV Dataset (Pandas DataFrames)	The HDI dataset is stored in CSV format and managed using Pandas. Since the project is prediction-based and does not require persistent user data storage, a relational database is not necessary.
4	**Cloud / Hosting / Deployment**	Flask Local Server / Render (or PythonAnywhere)	Flask's built-in server is used during development, while Render or PythonAnywhere can be used for deployment because they are easy to configure, cost-effective, and support Python web applications.

5	**Version Control & CI/CD**	Git, GitHub	Git is used for version control, and GitHub enables collaborative development, code management, issue tracking, and project sharing among team members.
6	**Third-Party APIs / Other Tools**	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Anaconda, Jupyter Notebook	Scikit-learn is used to build and train the machine learning model. Pandas and NumPy handle data preprocessing and numerical computations. Matplotlib and Seaborn generate visualizations. Anaconda provides the development environment, and Jupyter Notebook is used for model development and experimentation.